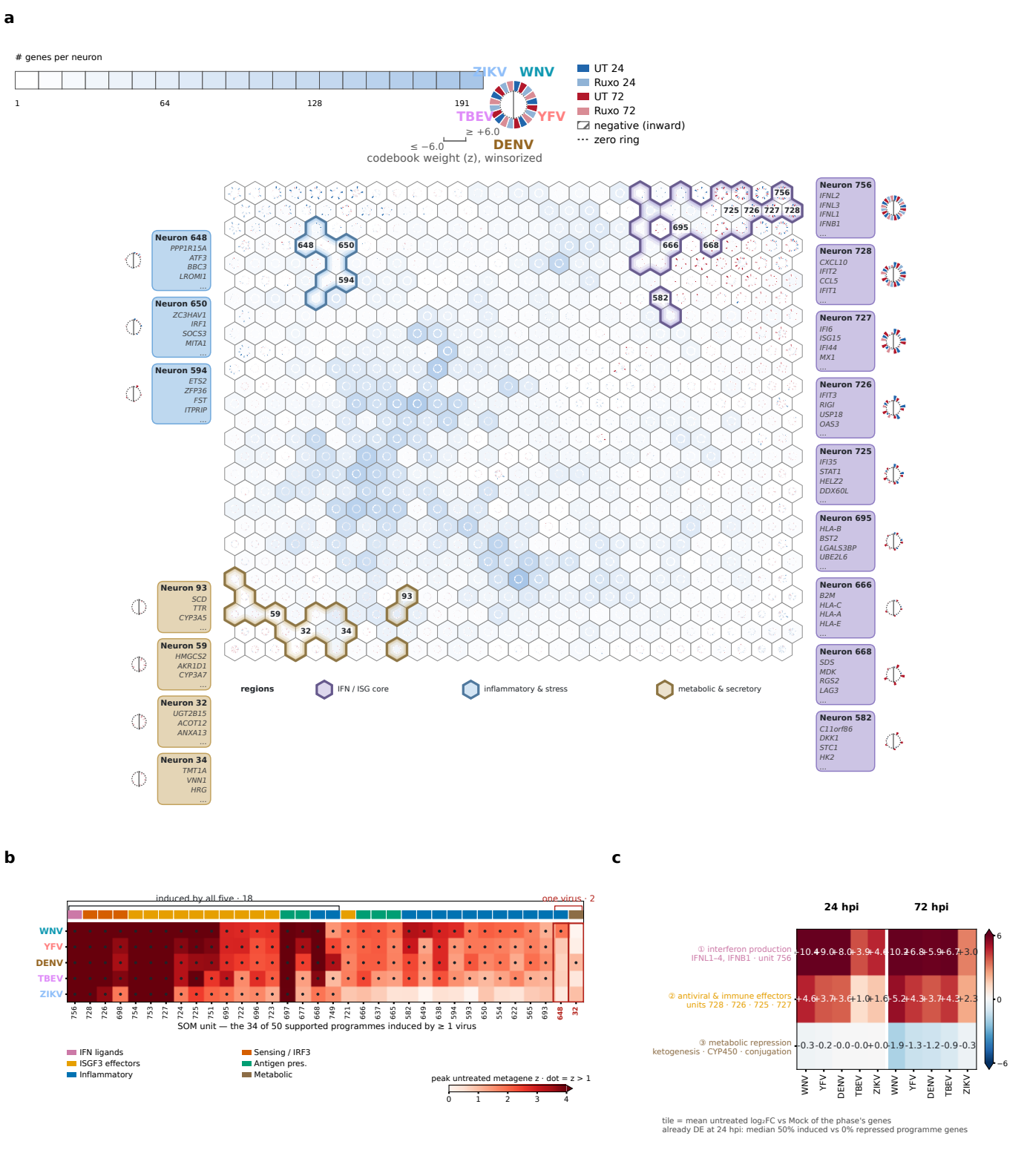

a
genes per neuron
1
64
128
191
ZIKV
WNV
TBEV
YFV
DENV
≥ +6.0
≤ −6.0
codebook weight (z), winsorized
UT 24
Ruxo 24
UT 72
Ruxo 72
negative (inward)
zero ring
Neuron 648
PPP1R15A
ATF3
BBC3
LROMI1
Neuron 650
ZC3HAV1
IRF1
SOCS3
MITA1
Neuron 594
ETS2
ZFP36
FST
ITPRIP
Neuron 93
SCD
TTR
CYP3A5
Neuron 59
HMGCS2
AKR1D1
CYP3A7
Neuron 32
UGT2B15
ACOT12
ANXA13
Neuron 34
TMT1A
VNN1
HRG
Neuron 756
IFNL2
IFNL3
IFNL1
IFNB1
Neuron 728
CXCL10
IFIT2
CCL5
IFIT1
Neuron 727
IFI6
ISG15
IFI44
MX1
Neuron 726
IFIT3
RIGI
USP18
OAS3
Neuron 725
IFI35
STAT1
HELZ2
DDX60L
Neuron 695
HLA-B
B2M
LGALS3BP
UBE2L6
Neuron 666
B2M
HLA-C
HLA-A
HLA-E
Neuron 668
SDS
MDK
RGS2
LAG3
Neuron 582
C11orf86
DKK1
STC1
HK2
regions
IFN / ISG core
inflammatory & stress
metabolic & secretory
b
induced by all five · 18
one virus · 2
WNV
YFV
DENV
TBEV
ZIKV
SOM unit — the 34 of 50 supported programmes induced by ≥ 1 virus
IFN ligands
ISGF3 effectors
Inflammatory
Sensing / IRF3
Antigen pres.
Metabolic
peak untreated metagene z · dot = z > 1
c
24 hpi
72 hpi
① interferon production
IFNL1–4, IFNB1 · unit 756
-10.4 -9.0 -8.0 -3.9 +4.0 -10.2 -6.8 -5.9 -6.7 -3.0
② antiviral & immune effectors
units 728 · 726 · 725 · 727
+4.6 +3.7 +3.6 +1.0 +1.6 -5.2 +4.3 +3.7 +4.3 +2.3
③ metabolic repression
ketogenesis · CYP450 · conjugation
-0.3 -0.2 -0.0 -0.0 +0.0 -1.9 -1.3 -1.2 -0.9 -0.3
WNV
YFV
DENV
TBEV
ZIKV
tile = mean untreated log₂FC vs Mock of the phase's genes
already DE at 24 hpi: median 50% induced vs 0% repressed programme genes